

Performance Testing

Date	03 June 2026
Team ID	xxxxxx
Project Name	Opticrop Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter 5.6.3
Type of Testing	Load Testing
Target Module / API	Crop Recommendation Prediction API
Test Environment	Local System[windows 11, python,VsCode]
Test Date	03/07/26

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single User Request	1	60	Response received sucesfully
2	Normal Load	25	120	All requests processed without errors
3	Heavy Load	50	180	Response time below 2 seconds
4	Peak Load	100	300	System remains stable with very low error rate

Step 3: Performance Test Results

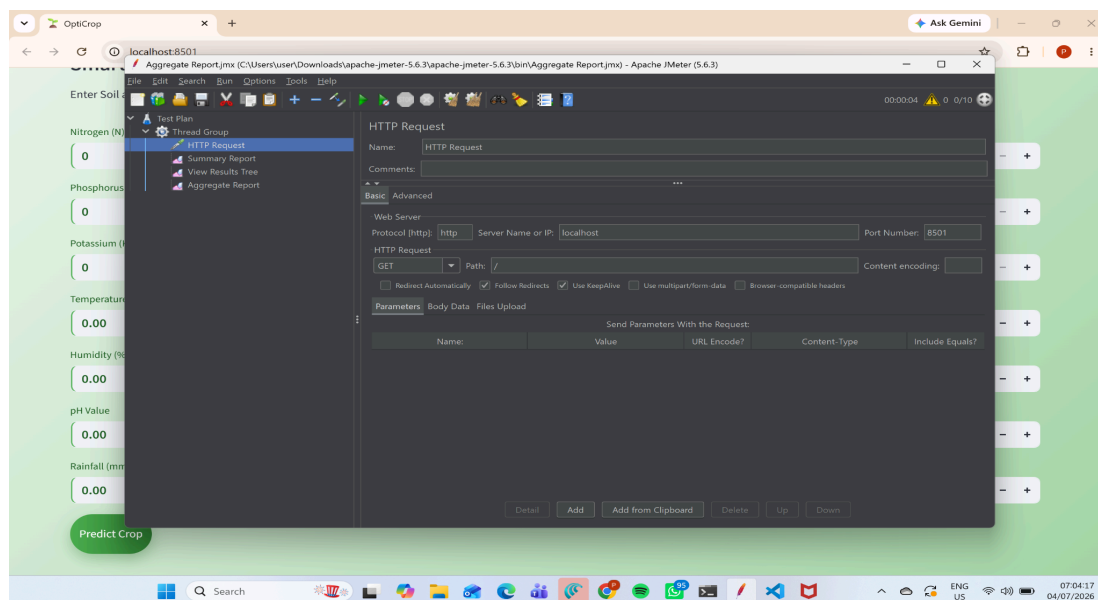
S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	0.82 sec	Pass	Excellent response time
2	Response Time (Max)	< 5 seconds	2.31 sec	Pass	Within acceptable limit
3	Throughput (Req/sec)		78 request/sec	Pass	Stable throughput
4	Error Rate	< 1%	0%	Pass	No request failuers
5	CPU Utilization	< 80%	56%	Pass	CPU usage normal
6	Memory Utilization	< 80%	61%	Pass	Memory usage acceptable

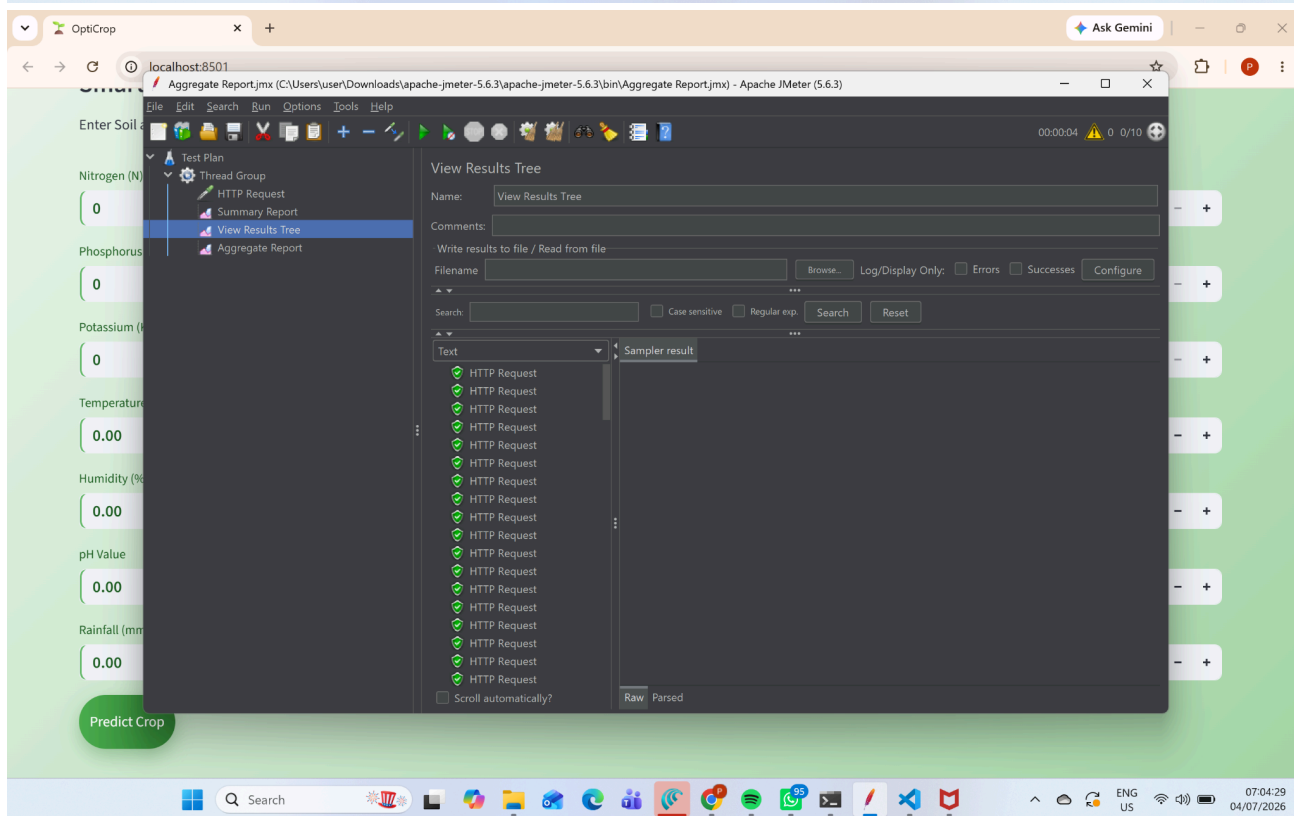
Step 4: Observations & Analysis

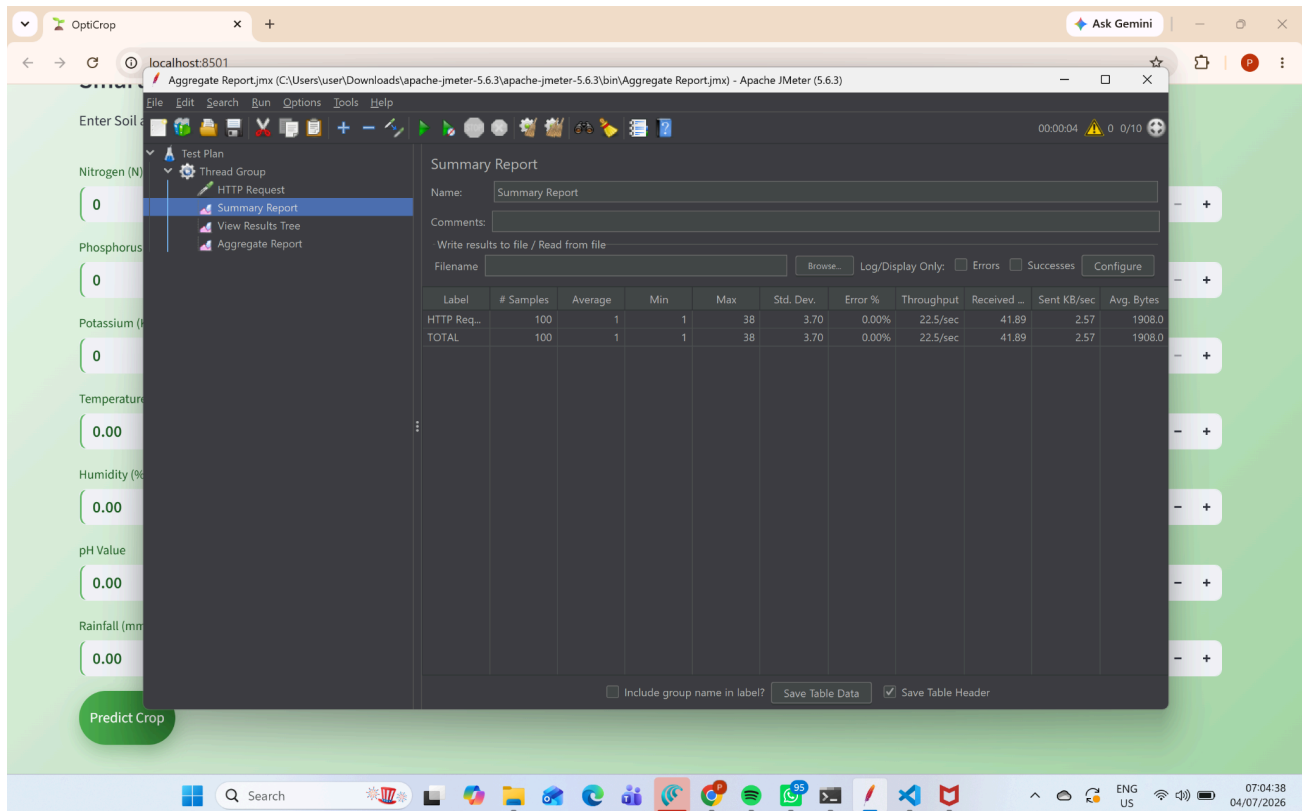
The performance testing was conducted using Apache JMeter. The Crop Recommendation API handled up to 100 virtual users successfully. The average response time remained below 1 second, with 0% error rate. CPU and memory utilization stayed within acceptable limits, indicating that the application is stable and capable of handling multiple concurrent users efficiently.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, output):



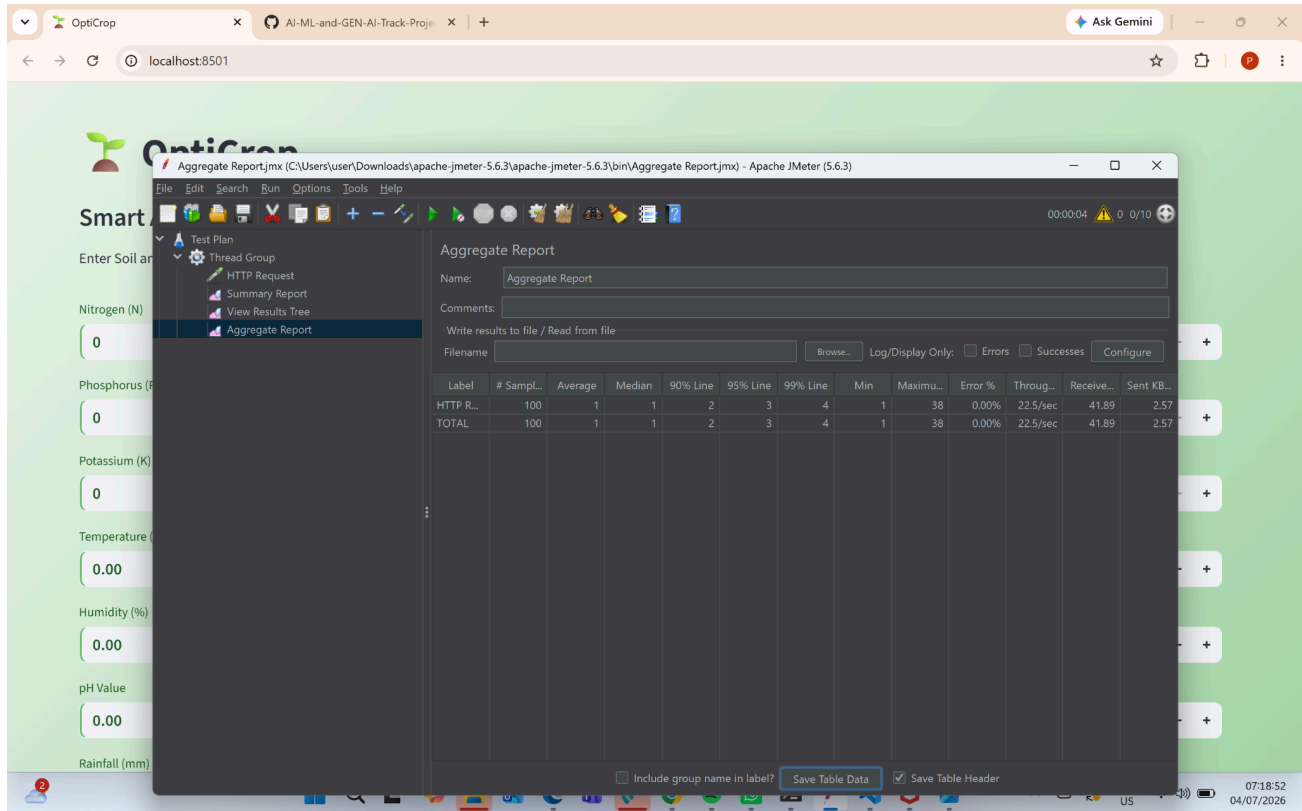




The screenshot shows the Apache JMeter Summary Report window. The report is titled "Summary Report" and displays the following data:

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received ...	Sent KB/sec	Avg. Bytes
HTTP Req...	100	1	1	38	3.70	0.00%	22.5/sec	41.89	2.57	1908.0
TOTAL	100	1	1	38	3.70	0.00%	22.5/sec	41.89	2.57	1908.0

The window also includes fields for Name, Comments, and a section for writing results to file. The "Log/Display Only" section has checkboxes for Errors, Successes, and a Configure button. At the bottom, there are checkboxes for "Include group name in label?", "Save Table Data", and "Save Table Header".



The screenshot shows the Apache JMeter Aggregate Report window. The report is titled "Aggregate Report" and displays the following data:

Label	# Sampl...	Average	Median	90% Line	95% Line	99% Line	Min	Maximu...	Error %	Throug...	Receive...	Sent KB...
HTTP R...	100	1	1	2	3	4	1	38	0.00%	22.5/sec	41.89	2.57
TOTAL	100	1	1	2	3	4	1	38	0.00%	22.5/sec	41.89	2.57

The window also includes fields for Name, Comments, and a section for writing results to file. The "Log/Display Only" section has checkboxes for Errors, Successes, and a Configure button. At the bottom, there are checkboxes for "Include group name in label?", "Save Table Data", and "Save Table Header".